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**TOPIC MODELING AND SENTIMENT ANALYSIS:
ANALYZING CULTURAL DIFFERENCES IN MOVIE
REVIEWS**

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SENIOR PROJECT

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LIST OF ABBREVIATIONS

BERT	Bidirectional Encoder Representations from Transformers
BERTopic	Topic Modeling technique based on BERT embeddings
CNN	Convolutional Neural Network
IMDb	Internet Movie Database
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
mBERT	Multilingual Bidirectional Encoder Representations from Transformers
NLP	Natural Language Processing
pLSA	Probabilistic Latent Semantic Analysis
RNN	Recurrent Neural Network
RoBERTa	Robustly Optimized BERT Pretraining Approach
TF-IDF	Term Frequency-Inverse Document Frequency

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ABSTRACT

TOPIC MODELING AND SENTIMENT ANALYSIS: ANALYZING CULTURAL DIFFERENCES IN MOVIE REVIEWS

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Senior Project

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The main subject of this project is making a comparative analysis of the movie reviewing styles of different cultures. To represent these cultures, viewers who review movies in Turkish and English on various platforms were selected. Firstly, for this purpose, Turkish movie reviews were collected from platforms whose primary language is Turkish, and English movie reviews were collected from platforms whose primary language is English. These comments were collected with appropriate web scraping methods. Then, the Turkish and English data were given into the BERTopic algorithm, which can process multiple languages at the same time, and separated into their related topics. The obtained topics labeled manually to make them suitable for analyzing with human eye. Later, the sentences from these separated topics were given into RoBERTa and Qwen2.5-7B models, which were trained with various datasets for Turkish and English languages separately. And the sentiment percentages of each topic were obtained. To confirm there is a relationship between the topics and their emotional distributions, chi-square independence test was applied. Finally, all of the results which were obtained in this project were analyzed with the tables and graphs.

Keywords: Turkish and English Movie Reviews, Topic Modeling, Sentiment Analysis, BERTopic, RoBERTa, Qwen, Chi-Square Test

KONU MODELLEMESİ VE SENTİMENT ANALİZİ: FİLM YORUMLARINDA KÜLTÜREL FARKLILIKLARIN ANALİZİ

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Bilgisayar Mühendisliği Bölümü

Bitirme Projesi

Danışman: Dr. Öğr. Üyesi Ayşe ÖCAL

Bu projenin ana konusu farklı kültürlerin filmlere yorum yapma stillerinin karşılaştırmalı analizini yapmaktır. Bu kültürleri temsil etmek için farklı platformlardaki Türkçe ve İngilizce dillerinde yorum yapan izleyiciler seçilmiştir. Bunun için öncelikle ana dili Türkçe olan platformlardan Türkçe film yorumları, ana dili İngilizce olan platformlardan ise İngilizce yorumlar toplanmıştır. Bu yorum toplama işlemleri uygun web kazıma yöntemleri ile yapılmıştır. Daha sonra elde edilen bu Türkçe ve İngilizce veriler, aynı anda birden fazla dil ile işlem yapabilen BERTopic algoritmasına verilerek konularına ayrılmıştır. Elde edilen konuların isimleri elle etiketlenerek insan gözüyle analiz edebilmek için uygun hale getirilmiştir. Sonrasında bu ayrılmış konulardaki cümleler, Türkçe ve İngilizce dilleri için çeşitli veri setleriyle ayrı ayrı eğitilmiş RoBERTa ve Qwen2.5-7B modellerine verilerek her konu için cümlelerin duygu yüzdeleri elde edilmiştir. Elde edilen konular ve konuların duygu dağılımlarının ilişkili olduğunu doğrulamak için ki-kare bağımsızlık testi yapılmıştır. Son olarak da bu projede elde edilen tüm sonuçlar, grafikler ve tablolar yardımıyla analiz edilmiştir.

Anahtar Kelimeler: Türkçe ve İngilizce Film Yorumları, Konu Modelleme, Duygu Analizi, BERTopic, RoBERTa, Qwen, Ki-Kare Testi

1

INTRODUCTION

In the globalizing digital world, cinema has evolved to a universal communication platform that represents norms, beliefs and sense of humor of people, rather than being only a visual source for entertainment. That means cinema is a great tool for different cultures to express themselves nowadays. This is why cinema is a significant field for understanding different cultures. Especially, reactions of people for well known movies provides the main source for analysing these differences between different cultures.

To understand better how cinema became this important, it is good to know the short history of cinema. The first ever recorded short movie "Roundhay Garden Scene" by French inventor Louis Le Prince in England in 1888 [1]. This short movie was just Le Prince's family members walking in the garden and it was only 2 seconds long. This is the moment that cinema entered to human life. And seven years later, on 28th of December, 1895, the first public movie "La Sortie de l'usine Lumière à Lyon" (Workers Leaving the Lumière Factory) screened by Lumière Brothers (Auguste and Louis Lumière) in Grand Café, Paris, France [2]. Since then, the technology has improved a lot, and today there are hundreds of movies released each year.

In this project, we aim to analyze how people discuss and how they feel after watching films. For achieving this, we divided this study into two parts: "Turkish-speaking viewers" and "global/non-Turkish-speaking viewers". For the analysis, films from the IMDb Top 250 list were taken as a reference. The data were collected via web scraping from IMDb and Letterboxd for English-language reviews [3] [4]. And for Turkish-language reviews; Beyazperde, Sinemalar, and Ekşi Sözlük was selected [5] [6] [7].

The collected data entered some preprocessing steps for making them suitable for this project. In the topic modeling stage, the multilingual BERTopic algorithm was used [8]. In the sentiment analysis stage, Qwen and RoBERTa models were fine-tuned on separate datasets for each language [9] [10]. In the final stage of the study, the

Chi-square test was applied to scientifically identify cultural differences between the distributions of topics and sentiments.

1.1 Aim of the Study

The main intention of this study is determining the main themes of audience reviews and analyzing the sentiment orientations of the themes. For doing this, the distribution between topics of Turkish and English reviews, and their clustering emotions are analyzed. And their identifying their differences is our main goal.

This method allows to examine the movie elements better such as cinematography, acting performance and ideological parts. Accordingly, this makes examining different focus points of different cultures much easier.

The main motivation of this study is contributing to multilingual natural language processing literature about movies. Most of the researchs are only about Western oriented reviews with the datasets which are only have English sentences. But this study has an aim to compare both Turkish and English reviews.

Rather than being a limited general sentiment classification, this study uses the topic-based sentiment analysis approach for making cultural comparison. Also, the effectiveness of Large Language Models (LLMs) in sentiment analysis tasks will be compared to classic encoder-based models. That makes this study is also aims to provide aid for computational linguistics and cultural sociology in different ways.

1.2 Preliminary Review

Nowadays, basic sentiment analysis tools are frequently used by digital streaming services and film production companies for evaluating people's feedback. However, current methods are mostly relying on word frequency-based techniques like TF-IDF or traditional dictionary-based approaches [11]. These systems generally determine the overall sentiment of a text with counting the emotionally strong words like "excellent," "perfect," "bad," or "terrible."

These systems are practical and fast. However, they are insufficient for analyzing texts that involve metaphors, irony, advanced level of structure usages, etc. And these are common in movie reviews. For example, a statement such as "The villain's performance was so terrifying that I couldn't take my eyes off the screen" may be classified as negative by a dictionary-based system. However, when we read this as a human, it is a positive sentence which doesn't have any negative meaning.

Review of similar academic studies and existing systems shows that most of the approaches either only performing general sentiment analysis for entire texts or only focusing on topic modeling. For example, there are systems that extract topics from IMDb reviews in a single language (either English or Turkish) using LDA (Latent Dirichlet Allocation) algorithms [12]. LDA-based methods only use word co-occurrence statistics. And this means they are unable to understand the true meaning of the words. It is unsuccessful in identifying if two words are related or not, unless they occur so much in the sentences. Also almost all of the systems that are currently used in studies are analyzing languages independently. And this means that this field isn't a commonly studied field for now.

This project makes use of balanced datasets gathered from international and Turkish film review platforms (IMDb, Letterboxd, Beyazperde, Sinemalar, and Ekşi Sözlük) to overcome these limitations. The study uses improved Qwen and RoBERTa models for each language and uses the multilingual BERTopic algorithm to produce topic groups of sentences which are language independent. Also, this project used statistical techniques like the Chi-square test for making a sentiment distribution analysis for each topic. And this project seeks to address the limitations of earlier researchs.

2 LITERATURE ANALYSIS

In this section, previous studies about topic modeling, sentiment analysis and natural language processing (NLP) are examined. And also clustering and making sentiment analysis about movie reviews in different fields are examined.

2.1 Sentiment Analysis on Movie Reviews: A Deep Dive into Modern Techniques and Open Challenges

This study presents a deep examination of the development of sentiment analysis in movie reviews from traditional systems to modern LLMs and Transformer-based architectures. According to the other researchers, modern models (LLMs and BERT-based models) achieve quite high accuracy rates like 97–98%. Despite the development of the models, these studies show some issues that are currently exists in this field. Such as domain or temporal shifts, being able to understand long texts, sarcasm and irony detection [13].

One of the main limits which is mentioned in this study is representing different cultures with different languages and their styles with the current technology. The studies were selected to reference in this report mentions about these problems. Like mainly focusing on sentences in English language. This is directly related with the purpose of our research. Our study hopes to fill some of the gaps in multilingual and cross cultural sentiment analysis with comparing Turkish and English movie reviews. And doing this with using modern architectures like Qwen and RoBERTa to cross-lingual cultural differences.

2.2 A Natural-Language-Processing-Based Method for the Clustering and Analysis of Movie Reviews and Classification by Genre

This study suggests a natural language processing-based method to working with movie review dataset that we created with web scraping from websites like IMDb.

In this study, unsupervised learning algorithms like K-means and Mini-batch K-means was used to group reviews into relevant categories.

Also, one of the main results of this study is the preprocessing steps are really important for determining performance of the models. The researchers claims that using TF-IDF for text vectorization and removing stopwords by using a threshold-based custom algorithm can result increasing the accuracy values up to 94% when matching reviews with their related movie genres [14].

The web scraping and topic modeling stages of our project have a strong methodological base thanks to that work. Similarly, it is expected that our dataset which was created from IMDb, Letterboxd, Beyazperde, Sinemalar, and Ekşi Sözlük with web scraping will be naturally noisy. This study emphasises the importance of preprocessing, including removing stopwords specific to both Turkish and English. This step helps to the multilingual BERTopic algorithm to focus on significant themes (such as acting and cinematography) rather than unnecessary words that can be used so often in a specific language.

2.3 A Review of Multilingual Sentiment Analysis

In this study, the development of multilingual sentiment analysis over the last ten years is examined. It covers topics like machine learning, deep learning, and approaches of the community. The study shows that while deep learning models (CNNs, RNNs, etc.) and Transformer-based architectures (mBERT, XLM-RoBERTa, etc.) outperform classical approaches when finding out the complex lingual patterns across languages. But also the classical machine learning models (Naive Bayes, Support Vector Machines, Random Forest, etc.) combined with TF-IDF or Bag-of-Words representations can achieve quite strong performance in specific languages.

Some studies also emphasises few important issues in multilingual sentiment analysis like the difficulty of studying with unstructured, informal text data and they also mention about the language-dependent performance variations [15]. The modeling approach used in our study is directly supported by the issue of performance differences in each language. The Qwen and RoBERTa models in our study are fine-tuned independently on language specific datasets. Because a model that performs well in English language does not have to achieve similar results in Turkish language.

Additionally, our study wants to create more unique result, because significant amount of the current literature is still only about the basic sentiment classification (such as

positive, negative, and neutral). With our approach, the reactions in different cultures can be understood better and more context oriented. With our approach, the reactions in different cultures can be understood more detailed and more context oriented.

2.4 Other Studies About Clustering and Analyzing Sentences

There are various studies about examining mentioned topics and existing sentiments in each English, Turkish and other languages. For example in an article written in 2025 named "BERTopic for Topic Modeling of Hindi Short Texts: A Comparative Study" mentions about how BERTopic algorithm is better at topic classification than standard probabilistic models such as LDA (Latent Dirichlet Allocation) and pLSA (Probabilistic Latent Semantic Analysis), which are traditionally accepted in the topic modeling literature [16] [17]. The researchers made this study in Hindi, which shows BERTopic algorithm can be used in languages other than English [18].

In another article, which is from 2021, named "An Application of Emotion Detection in Sentiment Analysis on Movie Reviews", emotion classification in movie reviews with a neural network created in Keras library [19]. This study used movie reviews in Filipino language (Tagalog) which is structurally different than English like Turkish. Four different emotion types are used in this study which are Masaya (happiness), Malungkot (sadness), Pagkagalit (anger) and Pagkatakot (fear). This is also an example about sentiment analysis in a different language than English [20].

An article from 2026 shows how traditional classification algorithms like Logistic Regression, Naive Bayes; Transformers based models like BERT and large language models (LLMs) like Llama classifies emotions in movie reviews [21]. The researchers says LLMs have more success than any other techniques. But they also mentions about the sentences with sarcasm can still be difficult to classify even for LLMs [22].

A different article from 2022, shows a study about how Turkish people make reviews about movies online in a detailed way. This study includes a part that has topic classification of movie reviews made in Turkish language [23]. This study doesn't involve any NLP, and we aim to achieve it with using BERTopic.

2.5 Contribution to the Literature

In past, topics and sentiments in the movie reviews of both English and Turkish comments have been analyzed by different researchers. But there is not any study compares both Turkish and English movie reviews by their topics and sentiments with

using natural language processing (NLP) tools. And also there aren't many studies in Turkish language, involves natural language processing (NLP) field while analyzing both Turkish and English reviews. So, this study aims to create a new source by using selected topic modeling and sentiment analysis tools to compare both Turkish and English movie reviews combined. And creating lists about Turkish and non-Turkish user review tendencies and feelings.

3.1 System Analysis

Our system is designed with five steps from data collection to statistical visualization. The system is designed to evaluate and compare sentiment and topic-based metrics gathered from audience reviews in various cultural contexts.

In the first step, multilingual movie reviews from movies in the IMDb Top 250 are collected. Turkish reviews are collected from Beyazperde, Sinemalar.com, and Ekşi Sözlük which represent local cinema culture. English language reviews are collected from international platforms which are IMDb and Letterboxd. For minimizing potential data imbalance problems later, the system uses web scraping techniques with Selenium and BeautifulSoup libraries to collect a balanced number of reviews in both languages for each movie.

In the second step, preprocessing and cleaning make sure that the raw data is changed into a format that is structured and noise free, which is what machine learning models need. We cleaned the texts we had collected by deleting their HTML tags, URLs, punctuation that doesn't matter, and expressions that only have numbers. Both languages also have some common stopwords, and those words that don't have any meaning were removed for preventing topic modeling results from being skewed.

In the third step, we found the dataset's fundamental thematic structure. A multilingual embedding model is used for vectorizing the cleaned and normalized data. The multilingual BERTopic framework reduces these embeddings into a useful low-dimensional space and then clusters them together to find important discussion topics like cinematography, acting, ideology, and narrative structure. Multilingual BERTopic makes it possible to align semantics across languages by putting similar concepts like "acting" and "oyunculuk" into the same semantic space.

In the fourth step, the sentimental analysis of sentences which are related to each identified topic is analyzed. Qwen and RoBERTa models are used for this step. These

models are designed to perform multi class sentiment classification and they are trained independently on Turkish and English language-specific datasets.

And finally, in the final step, the outcomes are provided for analysis. The extracted topic distributions and sentiment analysis are examined with Chi-square test. Then, the results are presented through visual schemes to make the analysis easier for both researchers and industry professionals.

Table 3.1 Performance and Evaluation Metrics Used in the Project

Metrics	Purpose
Accuracy	Ratio of correctly predicted reviews for RoBERTa and Qwen.
Precision	Ratio of correctly predicted positive predictions which are actually correct for RoBERTa and Qwen.
Recall	Ratio of actual positives which are correctly found for RoBERTa and Qwen.
F1-Score	Harmonic mean of precision and recall for class-imbalanced data.
Chi-square	Statistical significance of the topic-sentiment relationship.

3.2 Feasibility

The purpose of the feasibility analysis is to determine how resources will be used, how legal restrictions and financial limits will be handled and whether the project can be finished on schedule. This evaluation shows if the project can be completed successfully and achieve its goals.

3.2.1 Technical Feasibility

Technical feasibility involves evaluating the software and hardware infrastructure which is required for the project.

3.2.1.1 Software Feasibility

The project uses open-source libraries that are now standard in modern NLP research. Python (version 3.10+) is the main programming language in this project. Because it works well with many data analysis, machine learning, and NLP frameworks.

The development environment includes Windows and macOS operating systems, depending on the team members' preferences. Visual Studio Code is chosen as an

integrated development environment. Because it is flexible, can be expanded, and widely used by developers.

3.2.1.2 Hardware Feasibility

GPU memory capacity is one of the main bottlenecks in NLP projects, especially when fine-tuning LLMs.

Table 3.2 GPU Memory (VRAM) Requirements of Models Used in the Project

Model	Parameters	Method	Min. VRAM Requirement
RoBERTa-Base	~125 Million	Full Fine-Tuning	~2 GB – 6 GB
RoBERTa-Base-Turkish	~125 Million	Full Fine-Tuning	~2 GB – 6 GB
Qwen-7B	7 Billion	QLoRA (4-bit)	~20 GB – 24 GB

When we consider the sizes of our datasets, at least a system with 16 GB of VRAM is needed to efficiently train both RoBERTa and Qwen models. Although model training steps are performed with cloud-based resources like Google Colab, preprocessing steps are performed on our local machines, because the standard computers may not be able to meet those requirements.

Table 3.3 Personal Computer Hardware Features

Feature	Melih Yelman’s Computer	Faruk Altun’s Computer
Brand	Apple MacBook Air M3 2023	Lenovo Ideapad Gaming 3
Processor	Apple M2	AMD Ryzen 5600H
Graphics Card	Apple M2	Nvidia RTX3060
RAM	16 GB	16 GB
Storage	512 GB	512 GB
Operating System	macOS	Windows 10

3.2.2 Legal Feasibility

The web scraping processes planned in this project have been evaluated in terms of copyright law, database protection regulations, and data privacy considerations. Platforms such as IMDb, Letterboxd, Sinemalar.com, Beyazperde, and Ekşi Sözlük explicitly restrict automated data collection and the creation of commercial databases with their terms of service and robots.txt policies.

However, in both academic literature and common practice, data collection for

non-commercial research purposes is often considered as “fair use”. In this project, ethical and legal risks are minimized by not passing the technical constraints such as rate limiting, avoiding server overload, and anonymizing the collected data. And the data will be used only for non-commercial research and analysis purposes.

3.2.3 Economic Feasibility

The primary cost of the project comes from cloud-based services required for model fine-tuning. Google Colab is selected as the main platform for this purpose and the other stages of the project will be performed on our local machines.

Table 3.4 Economic Feasibility Analysis

Item	Quantity	Unit Price	Total Cost
Apple MacBook Air M2 2023	1	48,000 TL	48,000 TL
Lenovo Ideapad Gaming 3	1	19,000 TL	19,000 TL
Monthly Google Colab Pro Subscription	3	160 TL	480 TL
Total Project Cost			67,480 TL

3.2.4 Workforce and Timeline Feasibility

For achieving the project objectives, a structured timeline and efficient planning are required. The project workflow is designed as a sequence of interdependent stages and is planned to be completed in a 15-week schedule.

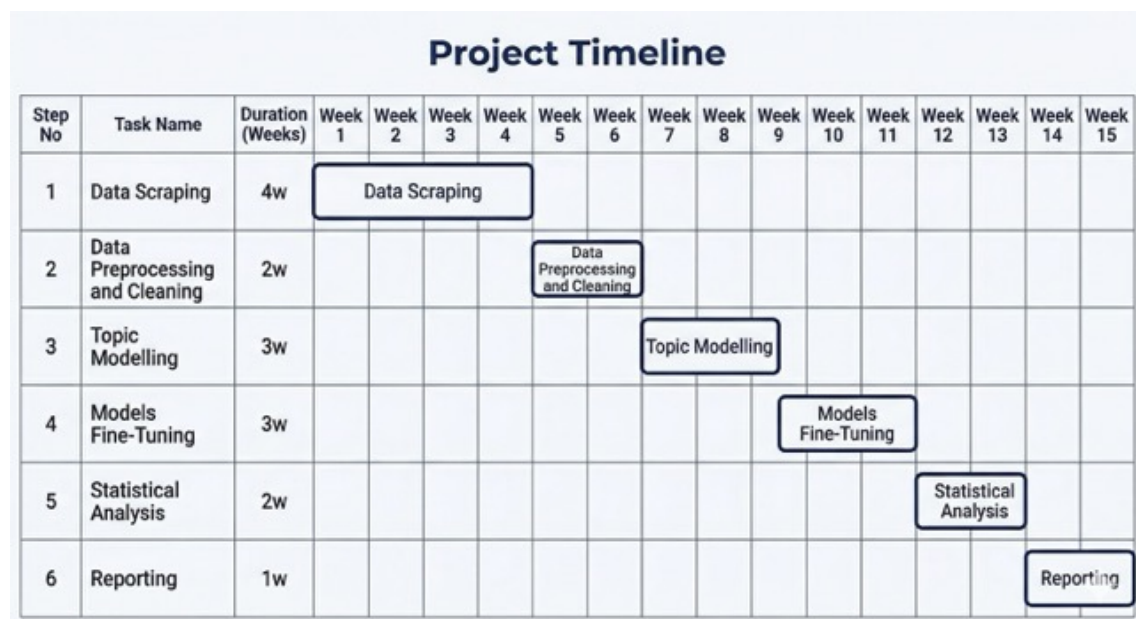


Figure 3.1 Gantt Chart

4 SYSTEM DESIGN

In this section, overall structure of the project, decisions about creating the datasets and the algorithms used are introduced. The system consists of six main modules: data collection, data preprocessing and balancing, topic modeling, sentiment analysis, and statistical comparison. The flow between the modules is shown in the block diagram.

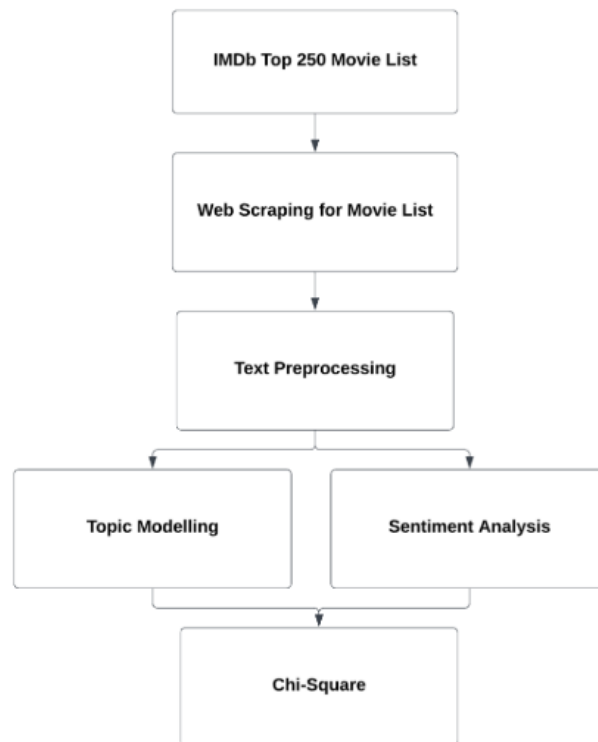


Figure 4.1 System Block Diagram

4.1 Material and Dataset

In this section, generally discussed about dataset, programming languages, models and libraries used in the project.

4.1.1 Movie Review Dataset

The study is based on films from IMDb’s Top 250 list. This list consists of 250 films which are considered popular worldwide, and is regularly updated based on IMDb user ratings. Five online platforms were selected to gather a balanced number of reviews in two languages for each film.

Table 4.1 Used Websites for Movie Reviews

Platform	Language	Rationale
IMDb	English	The most established film criticism and review platform.
Letterboxd	English	Young and active moviegoers
Beyaz Perde	Turkish	The most established cinema platform in Turkey
Sinemalar	Turkish	Another Turkish cinema platform
Ekşi Sözlük	Turkish	Platform for Turkish cultural criticism.

The decision to use two sources for English and three for Turkish is a design choice aimed at reducing stylistic bias specific to a single platform.

4.1.1.1 Data Scraping

Reviews were collected via web scrapers which are customized for each platform. All scrapers are designed according to the following common principles:

- Ethical scraping: Randomly spaced delay between requests; rotating User-Agent pool.
- Checkpoint logic: Permanently writing the result when each movie is completed; no restarting after interruptions.
- Asynchronous browser usage: Browser emulation was used for platforms with bot protection.
- Each review is stored along with the IMDb ID of the movie, the source platform, and raw text fields.

4.1.1.2 Data Preprocessing

The collected raw comments are converted into two versions. One for preserving the functionality of transformer-based models on natural texts and other one to provide a cleaned-up version for word-based analyses.

Table 4.2 Data Processing Steps

Process	Rationale
Unicode NFC normalization	To prevent incorrect word breaks caused by letters such as “İ”
Turkish sensitive lowercase conversion	For example, applying the transformation İ→i, I→ı in the correct order
Delete HTML tag and URL	Cleaning up marking tag noise from the platform
Delete non-letter characters	Clearing numbers, punctuation and special symbols
Stopword filtering	NLTK default lists + manual TR functional words + movie/cinema domain words
Lemmatization	Arrangement of the agglutinative structure of Turkish (e.g. films / film / film → film)
Deleting duplicate reviews	Deleting duplicate reviews
Word count threshold	Word-based threshold instead of character count.

After these steps, each review is stored in two separate text versions:

- Raw text version: It is used for the embedding phase of topic modeling and sentiment analysis models. Transformer architectures work with their own tokenizers. Punctuation, capitalization, and negation words are preserved because they carry semantic signals.
- Cleaned and lemma version: c-TF-IDF is used for topic word extraction and descriptive word frequency analyses. By reducing the dictionary size, word-based methods are enabled to produce more meaningful tags.

4.1.1.3 Stopwords and Lemmatization

During preprocessing, the stopwords lists were designed as a two-tiered extension of the standard NLTK sets:

- General functional words: The list suggested by NLTK for Turkish is supported by additional frequent functional words such as bir, sonra, ama, olan, olarak.
- Domain-specific stopwords: Since the project domain is movie review, domain words that are frequently seen in each review, such as film, sinema, izle, oyuncu,

yönetmen, sahne, karakter (TR) and movie, film, watch, scene, character, actor, director (EN), have been added. Without this filtering, the topic model would generate the same domain words as tags for each set.

Simplemma has been chosen for lemmatization. This choice ensures the portability of the system and its ability to operate with a single call signature in a common Turkish-English pipeline.

4.1.1.4 Film Based TR/EN Compensation

The main hypothesis of the project, which is the effect of cultural differences on the comment content, requires the same number of Turkish and English comments per movie. Otherwise, the observed differences can occur because of sample imbalance, not the language itself.

Therefore, the balancing strategy described in Algorithm 1 was developed.

Algorithm 1 Dataset Balancing Strategy

```

1: Constants:  $MIN\_REVIEWS = 100$ ,  $THRESHOLD = 0.05$ 
2: for all movie  $m$  in dataset do
3:    $n_{tr} \leftarrow$  Number of Turkish reviews in  $m$ 
4:    $n_{en} \leftarrow$  Number of English reviews in  $m$ 
5:   if  $n_{tr} < MIN\_REVIEWS$  or  $n_{en} < MIN\_REVIEWS$  then
6:     Remove movie  $m$  from the dataset
7:     continue
8:   end if
9:    $rel \leftarrow \frac{|n_{tr} - n_{en}|}{\max(n_{tr}, n_{en})}$ 
10:  if  $rel \leq THRESHOLD$  then
11:    All reviews in both languages remain in the dataset
12:  else
13:     $min\_count \leftarrow \min(n_{tr}, n_{en})$ 
14:    Select  $min\_count$  records randomly from the majority side
15:    The entire minority side remains
16:  end if
17: end for

```

4.1.1.5 Data Structure

Each review record includes the following fields: the film's IMDb ID, source platform, language tag, raw text version, and cleaned text version.

Number of reviews according to platforms are shown in Figure 4.2.

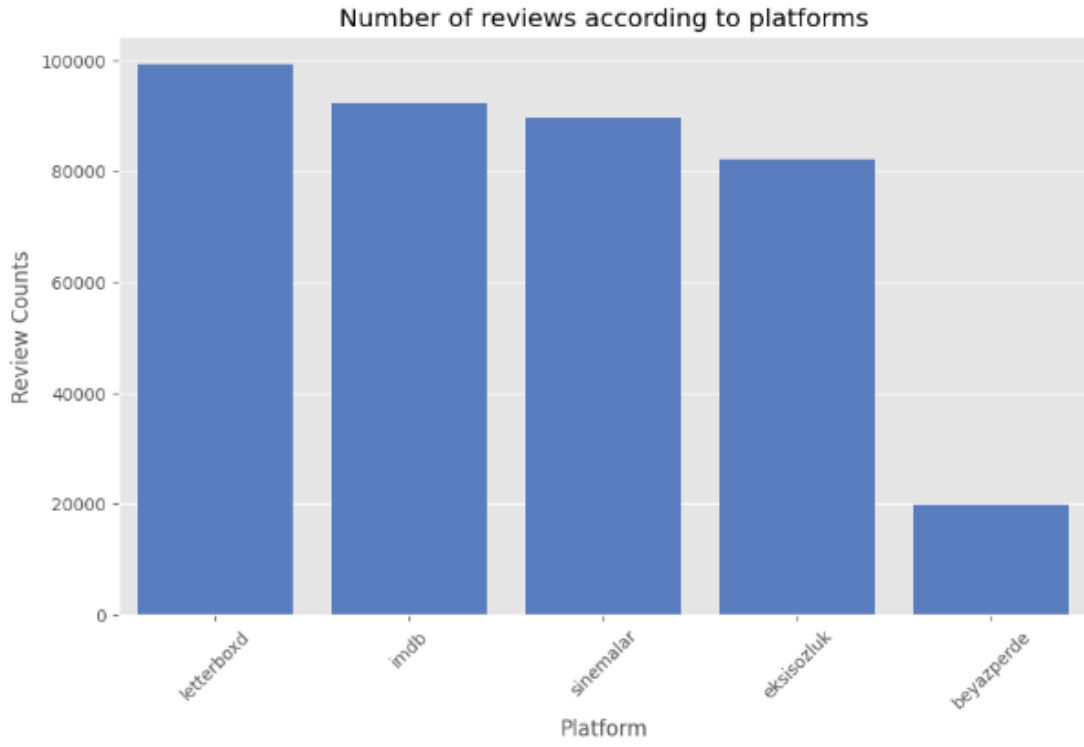


Figure 4.2 Number of Reviews According to Platforms

Word count distribution of reviews are shown in Figure 4.3.

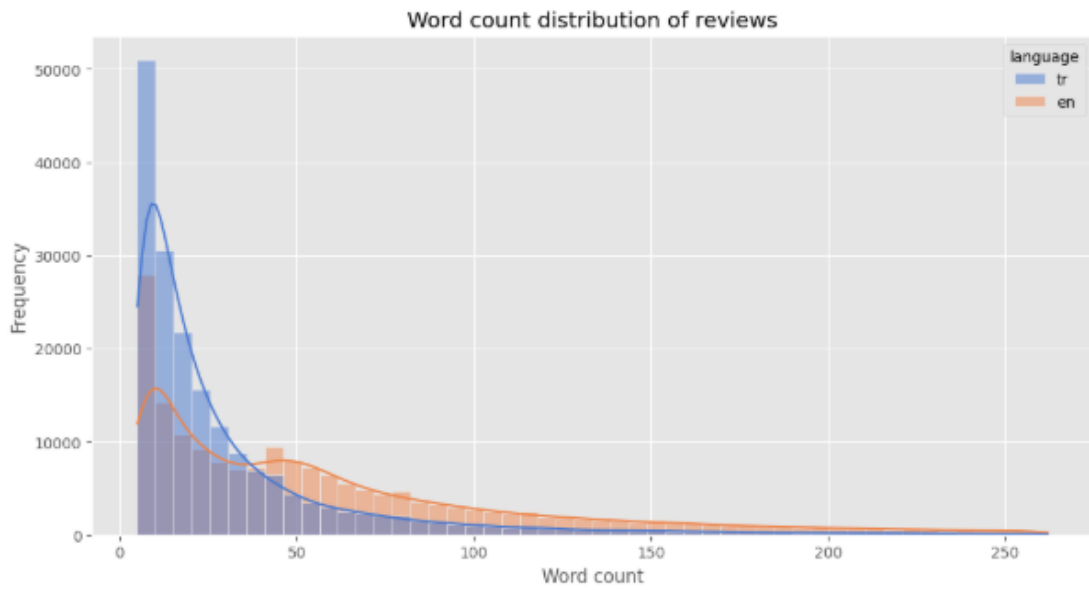


Figure 4.3 Word Count Distribution of Reviews

4.1.2 Emotion Classifying Dataset

In this section, the chosen datasets for emotion classification are explained with the related tools, materials and methods.

4.1.2.1 Structure of Datasets

Firstly, to be able to achieve a proper emotion classifier model, there is a dataset which contains sentences and their labeled emotions needed. We decided to use the relevant datasets about emotion classifying from the Internet. We created two different datasets, one for English and one for Turkish models.

We selected the well known GoEmotions dataset from Google for preparing English models [24]. The dataset contains 27 different emotions and we took only the sentences of selected emotions. And we selected TREMO dataset which is commonly used in Turkish sentiment analysis projects. Also the Turkish neutral labeled sentences were taken from Winvoker’s sentiment dataset [25]. The most relevant emotions that we might encounter most often in daily basis when reading movie reviews were selected. There are 6 emotions for each language. These emotions are anger, disgust, fear, happiness, neutral and sadness. For the English dataset, each emotion has 1500 sentences. And for the Turkish dataset, there is 2000 sentences for each emotion type. We preferred this for making finetuned models have stronger sense of understanding positive sentences rather than just being able to understand basic sentences related to happiness. Also, we didn’t preferred to use any lemmatization, cleaning and neglecting stopwords processes to the dataset because of the Transformers and LLM models consider the usage of stopwords and cased letters [26].

4.2 Method

This subsection presents the algorithmic components of the system, parameter selections, and referenced academic sources.

4.2.1 Topic Modeling

BERTopic, which is used for topic modeling, is a modular topic modeling algorithm created by Grootendorst in 2022 [8]. Unlike traditional LDA methods, it works on transformer-based sentence embedding vectors instead of word frequency distributions. So, texts written with semantically similar but different words can be placed on the same topic.

4.2.1.1 Sentence-Transformers

Each review is converted into a vector via a multilingual sentence embedding model. This project used the paraphrase-multilingual-MiniLM-L12-v2 model, which is built on the Sentence-BERT architecture, scaled down by MiniLM distillation, and trained on parallel sentence pairs in more than 50 languages [27] [28]. This model was selected primarily because of its multilingual capabilities, which are a prerequisite for multilingual BERTopic. By embedding both Turkish and English reviews into a shared vector space, the model ensures that cross-lingual comments discussing the same topic are positioned with each other. The input for the embedding phase is raw review text. This choice is made because the transformer models are trained on natural texts which are using their own tokenizers, punctuation, capitalization and negation words which are carrying critical information for the meaning.

4.2.1.2 UMAP

High-dimensional embedding vectors are reduced to a lower dimension. So, the clustering algorithms are not affected by the problem of dimensionality problem. The UMAP method is used for this process. The parameters used for the UMAP algorithm are provided in Table 4.3 [29].

Table 4.3 UMAP Algorithm Parameters

Parameter	Value
n_neighbors	30
n_components	5
min_dist	0
metric	cosine
random_state	42

4.2.1.3 HDBSCAN

HDBSCAN algorithm was used for clustering the reduced vectors. HDBSCAN automatically discovers irregular topic structures in the vector space based on data density without the need to predetermine the number of clusters [30]. And it improves the clustering quality by excluding outliers that do not belong to any topic with a -1 label instead of forcing them for putting them in a cluster. The parameters configured for the HDBSCAN algorithm are detailed in Table 4.4.

Table 4.4 HDBSCAN Algorithm Parameters

Parameter	Value
min_cluster_size	200
min_samples	10
metric	euclidean
cluster_selection_method	eom

4.2.1.4 c-TF-IDF

The class-based c-TF-IDF method was used to assign meaningful word tags to the created clusters. In this approach, all documents in each cluster are combined as a single document and cluster-based scores are calculated [8]. Thus, characteristic words that are frequently seen in the relevant cluster and rarely seen in other clusters. And the words with the highest scores are selected as the tag of that topic. The input for the c-TF-IDF stage is cleaned and lemmatized text. This choice is consistent with the natural behavior of word-based methods. If stopwords and agglutinative word inflections were not filtered, the same general words would be output as tags for every topic. The parameters configured for the c-TF-IDF process are detailed in Table 4.5.

Table 4.5 c-TF-IDF Algorithm Parameters

Parameter	Value
stop_words	TR+EN
min_df	50
max_df	0.95
ngram_range	(1, 2)

4.2.2 Emotion Classification

In this section, the system design is described as how it works and it's evaluation stages.

4.2.2.1 Pretrained Models

In this study there is 2 different type of language models used. First one is the classic encoder only type models of RoBERTa (Robustly optimized BERT approach). For the movie reviews in English language, roberta-base pretrained model is used. And for the movie reviews in Turkish language, roberta-base-turkish-uncased pretrained model created by Burak Aytan is used [31]. And the second one is the decoder only type models of Qwen family for both movie reviews in English and Turkish language.

4.2.2.2 Tools and Libraries

In this project, Python programming language is used for emotion classification. The libraries used for emotion classification are as follows:

- Transformers (HuggingFace) – Loading and fine-tuning RoBERTa and Qwen models.
- PyTorch – Implementing and fine-tuning the deep learning models.
- Pandas, NumPy – Data processing and analysis
- Matplotlib, Seaborn – Graphics and visualization
- scikit-learn – Model evaluation metrics (accuracy, precision, recall, F1-score)

4.2.2.3 Tokenization

In this project, the dataset sentences are processed using tokenizers compatible with the RoBERTa and Qwen models. Tokenization process is needed for turning sentences to numerical values. RoBERTa uses the RoBERTaTokenizer (BPE/WordPiece based) and Qwen models use the SentencePiece based tokenizer [10] [32]. During the tokenization, each text is limited to maximum of 64 tokens for controlling computational costs, time and having a fixed input size.

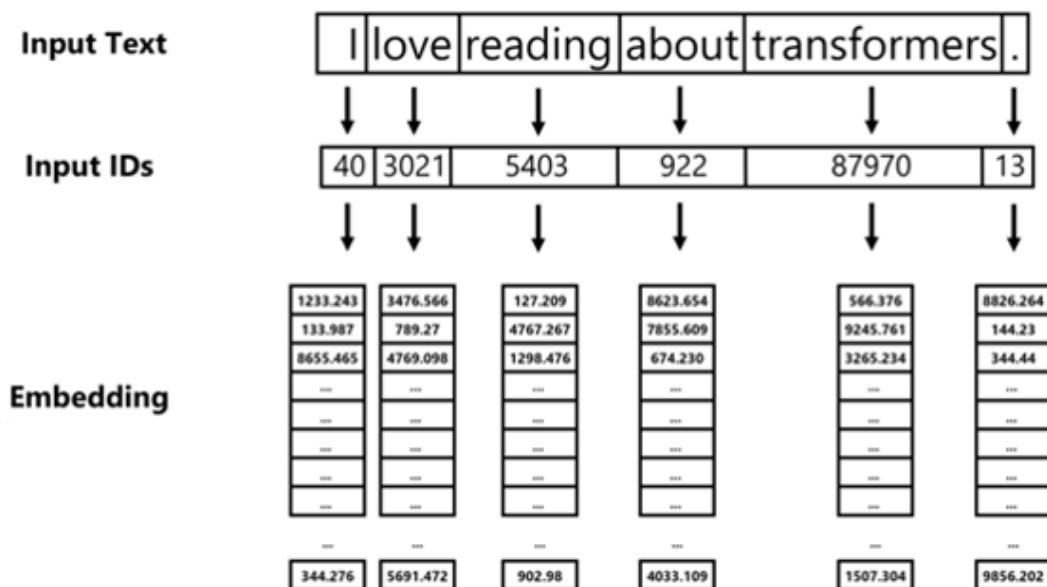


Figure 4.4 Tokenization Structure of Transformers Models

4.2.2.4 Fine-Tuning of Models

In the fine-tuning process, both pretrained RoBERTa and Qwen models are fine-tuned before the movie reviews are given into the model. The hyperparameters are used in model training for controlling the learning process of models. Example training parameters are as follows:

- Number of epoch: 1-3
- Learning rate: Between $1e-4$, $2e-5$
- Batch size: Between 4 - 32
- Optimizer: AdamW [33]
- Learning Rate Scheduler: StepLR with gamma variable: 0,5 for RoBERTa models (This variable is reduced after each epoch). Cosine / Linear decay for Qwen models.
- Loss Function: CrossEntropyLoss [34]
- Max sequence length: 128 tokens
- LoRA rank (r) (Only for Qwen Models): 8 / 16 / 32
- LoRA alpha (Only for Qwen Models): 16 / 32
- Dropout (Only for Qwen Models): 0.05 – 0.1
- Quantization (Only for Qwen Models) : 4-bit QLoRA

During the training process, validation loss and accuracy values are checked with TQDM library.

4.2.2.5 Evaluation

The model was evaluated both on the movie reviews dataset and on manually labeled reviews which are labeled by us.

The metrics used are:

- Accuracy: Overall success rate
- Precision: Accuracy of positive predictions
- Recall: Detection rate of true positives
- F1-Score: Harmonic mean of Precision and Recall

4.2.3 Chi-Square Test

Chi-Square test is made for examining the relationship between topic modeling and emotion classification [35]. The intention is finding a relationship if there is a significant relationship between some topics and specific emotions. For example, sentences of a specific topic which has negative or critical words may have higher possibility of being classified as an anger or disgust sentence. Another instance is, sentences of another specific topic which has positive and excitement words may have higher possibility of being classified as a happiness or surprise sentence.

5

EXPERIMENTAL RESULTS

In this section, the numerical results obtained from system components are presented and commented comparatively. The results were evaluated in three main parts. These are BERTopic topic modeling outputs, performance of RoBERTa and Qwen based sentiment classification models, and statistical validation of topic, sentiment and language relationships with the Chi-Square test.

The main corpus of study includes 229 movies from IMDb Top 250 movie list. It consists of 382,977 reviews in total. Dataset balancing strategy was used for getting close amount of reviews. The Turkish dataset have 191,415 different comments and the English dataset have 191,562 comments with this strategy.

The number of distribution of the reviews between five platforms is as follows: IMDb (96,969), Letterboxd (111,712), Sinemalar (103,133), Ekşi Sözlük (84,764), Beyaz Perde (20,967).

Table 5.1 Distribution of Reviews Across Platforms

Platform	Amount	Language
IMDb	96,969	English
Letterboxd	111,712	English
Sinemalar	103,133	Turkish
Ekşi Sözlük	84,764	Turkish
Beyaz Perde	20,967	Turkish

5.1 Performance Analysis

In this section, how different models achieve their results are shown. Success rates are determined with different performance metrics. And the existing studies examined for having an idea of the success rates of our models. These are explained in this section.

5.1.1 Performance Results of Models

The two different models (RoBERTa and Qwen) were fine-tuned separately on Turkish and English datasets for the same 6 emotion classification classes. These are anger, disgust, fear, happiness, neutral and sadness. Their performance results with metrics are shown in Table 5.2.

Table 5.2 Performance Results of Models

Model	Language	Accuracy	F1 Score (Weighted)
RoBERTa (roberta-base-turkish-uncased)	Turkish	0.91	0.91
RoBERTa (roberta-base)	English	0.78	0.77
Qwen 2.5 7B + LoRA	Turkish	0.94	0.94
Qwen 2.5 7B + LoRA	English	0.76	0.76

5.1.2 Evaluation of the Performance Results

The Qwen model achieved 3 percent higher F1 score than RoBERTa in Turkish. The reason of this difference probably because of Qwen’s higher number of parameters which is 7 billion and the LoRA adaptation from the instruction tuned base model. And the Turkish RoBERTa model is weaker since the base RoBERTa model was only trained in English.

Then, in English, the accuracy score of RoBERTa base model is nearly 1.5 percent is higher than the Qwen model. This maybe caused due to RoBERTa’s architecture fits better than Qwen’s in classification tasks. However, since the difference is so low, looking for a reason for this may not be that true. The performance in Turkish language is higher in both of the models, because of the TREMO dataset has shorter and much more precise sentences than the English GoEmotions dataset.

5.1.3 Evaluation of Other Studies

In a study, which is the main researchers of TREMO dataset testing it with Support Vector Machine Algorithm, they achieved an F1 score of %80 [25]. Another study used Mistral 7B model, which is very similar with Qwen 7B model, and ISEAR dataset achieved a %76 F1 score with 7 different emotions in English language and these scores are quite similar with our results [36].

The main study of GoEmotions dataset which used all 27 emotions of GoEmotions dataset achieved a %46 F1 score and %64 for only Ekman emotions [24]. Even though the class amount was reduced, the success rate is not as high as other studies because

of the model is fine-tuned with all 27 emotions at once.

A different study used various XLM family models in their study and fine-tuned them with a dataset that contains nearly 1 million texts in 23 different languages, including Turkish and English. Their texts have 11 different emotion classes and achieved an F1 score of %85.3 on average [37]. This study shows that an extremely high amount of samples can increase the success of models, since the more models encounter different structures, the more they can predict correctly.

In the another study which is made in Turkey, the author wanted from the middle school students to write small texts in 5 different emotions. And then they used TF-IDF with Turkish Lemmatization. And finally they achieved an average of %72 F1 score with LR, SVM algorithms and MLP, CNN architectures. They even mentioned about using BERT based models, but since their sample size is low, their outputs weren't eligible for this task. So they claimed that the classical techniques performs better with small sample sizes [38].

The examined articles shows that the success rates of our models are eligible for the emotion classification task.

5.2 Comparative Analysis

In this section, the comparisons between Turkish and English reviews are analyzed with tables and graphs which are created with results of the used models. And lastly, the Chi Square test results added for verifying the results are consistent.

5.2.1 BERTopic Topic Modeling Results

The results of BERTopic algorithm is analyzed in this part. Distribution of each topic, genres of the topics and the amount reviews in each language for the topics are analyzed. The analyzes were made with related graphs and tables in each part.

The multilingual paraphrase-multilingual-MiniLM-L12-v2 embedding + UMAP + HDBSCAN pipeline generated 239 clusters in the first pass. Merging with `reduce_topics(80)` and reassigning with `reduce_outliers` resulted in 80 meaningful topics. And the outlier rate is kept at only %0.15 (584 comments).

Topic sizes ranges between 504 and 55,531. The most crowded topics are Topic 0 with 55,531 reviews, then Topic 1 with 50,707 reviews, and finally topic 2 with 22,337 reviews. And these 3 topics represent %33 of all reviews. The most crowded 25 topics and their review amounts are shown in Table 5.1.

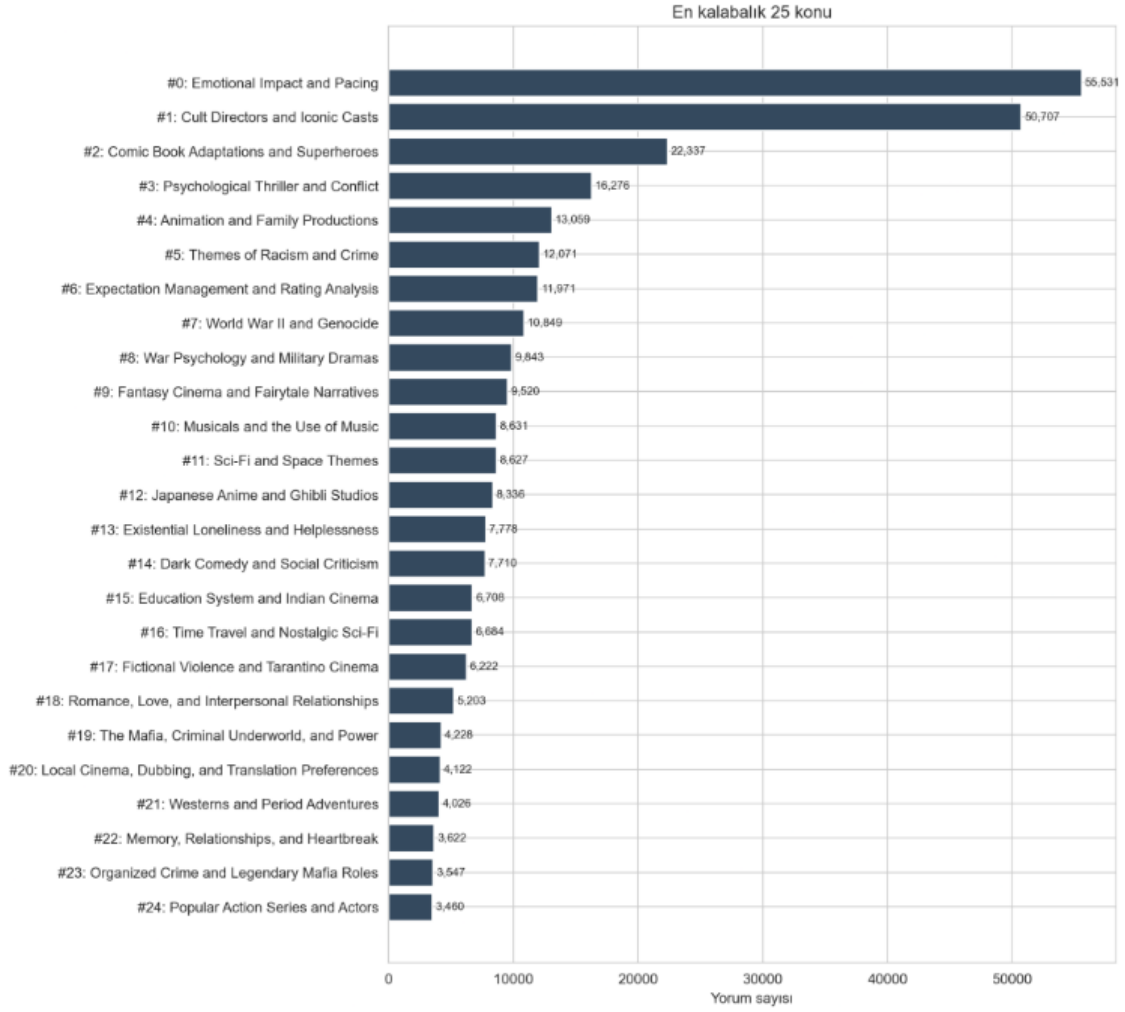


Figure 5.1 The Most Crowded 25 Topics

5.2.1.1 Topic Labeling and Categorization

The keywords and the sample documents were examined and easily readable titles were added to 80 topics. These topics categorized into 5 different categories. These categories are shown in Table 5.2.

Table 5.3 Distribution of Categories for the 80 Topics

Category	Topic Count
Genre Related	41
Social / Cultural Themes	16
Cast / Cinematography / Technical Elements	9
Audience Experience	7
Psychological Themes	6

For making healthier analyses, 23 topics which have balanced amount of reviews in both Turkish and English languages. The category based and topic based comparisons were made with these selected topics.

5.2.1.2 Topic and Language Distribution

The Turkish review rate of each topic is shown as bar graph in Figure 5.2.

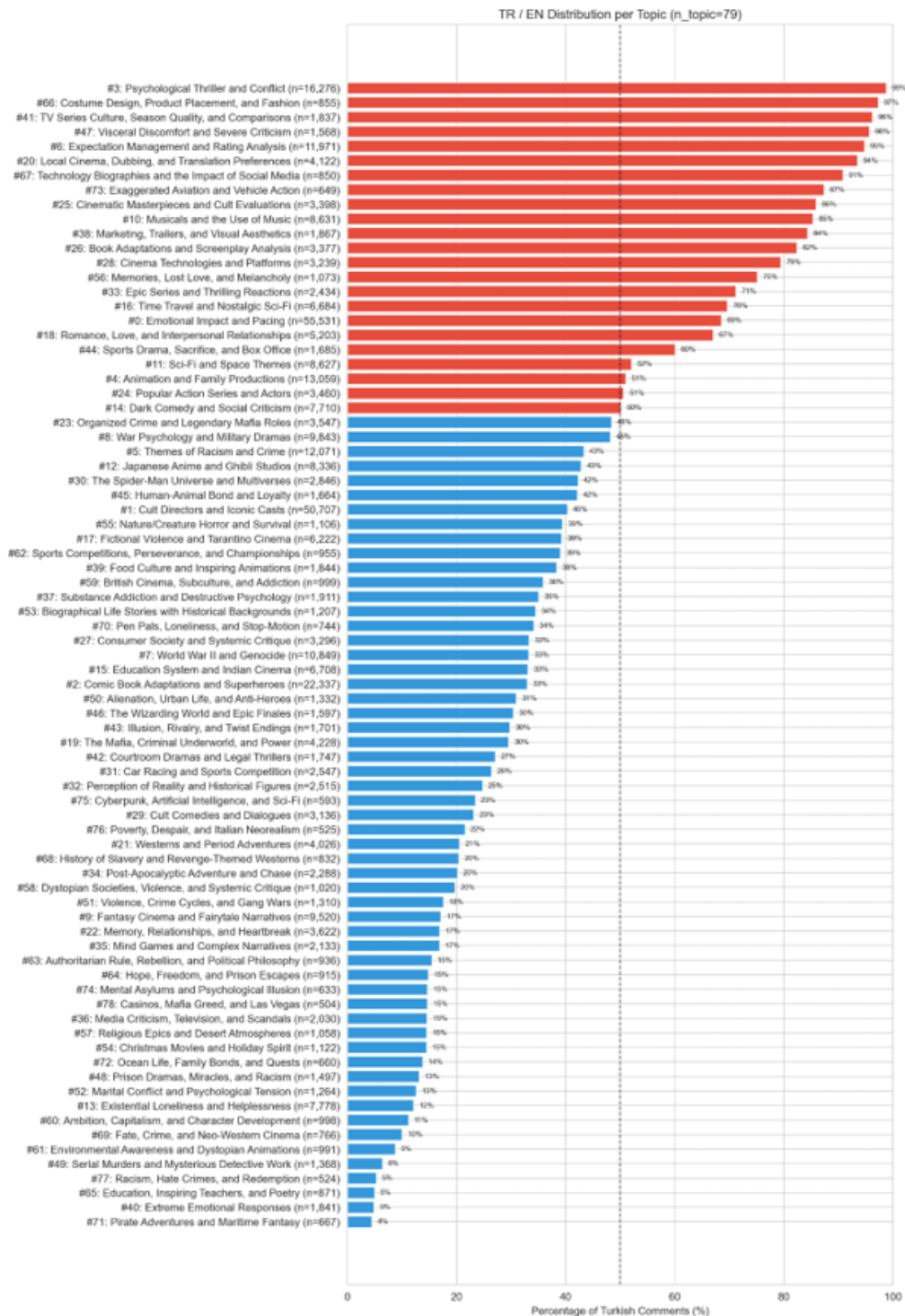


Figure 5.2 The Turkish Review Rate of Each Topic

The language distribution of the 23 filtered topics between Turkish and English is shown in Figure 5.3.

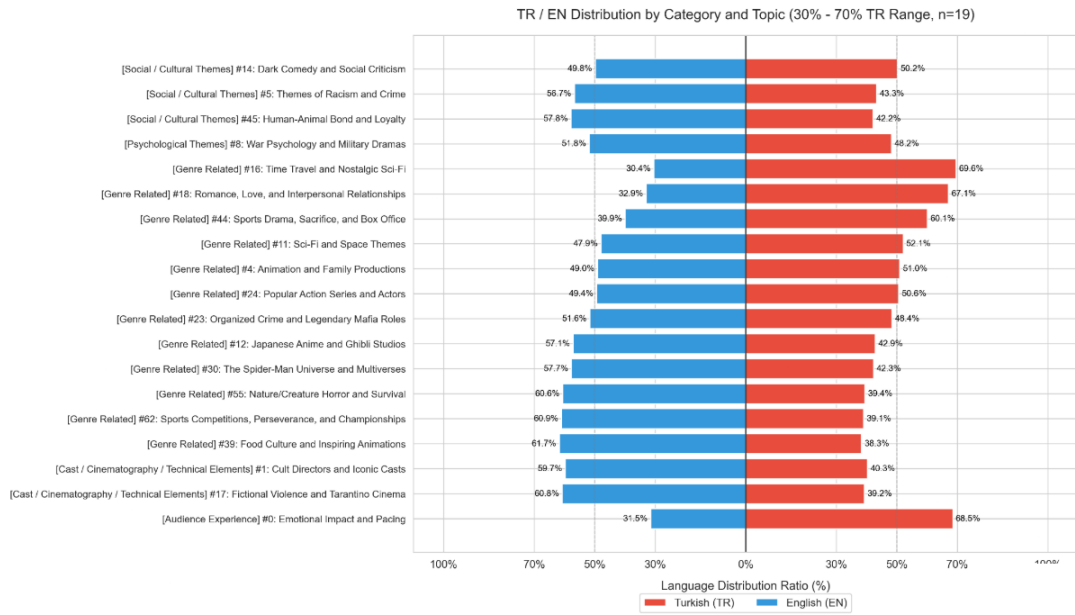


Figure 5.3 The Language Distribution of the 23 Filtered Topics

5.2.2 Compatibility Between Models

The predictive consistency performance of the two independent models for emotion classification (RoBERTa and Qwen) was measured with Cohen's Kappa score. The confusion matrix is presented in Figure 5.4.

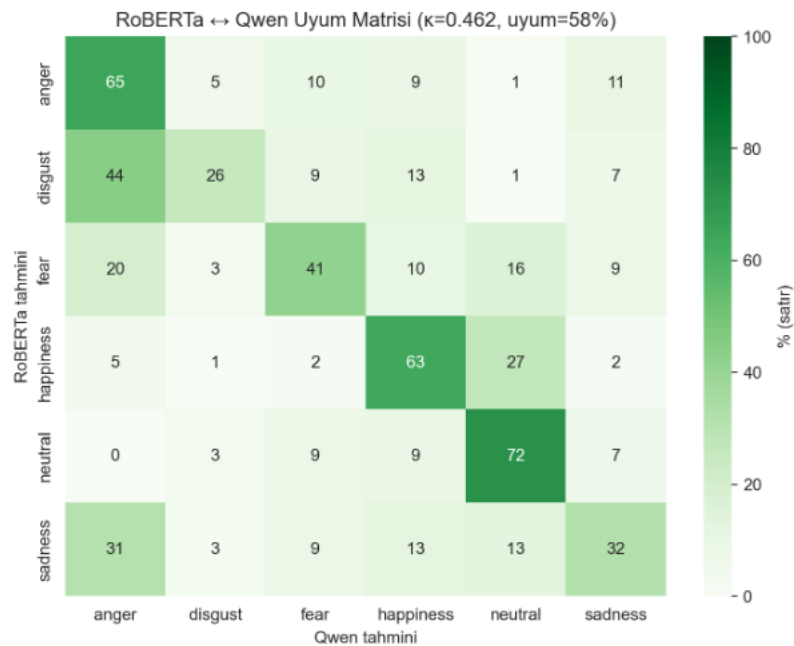


Figure 5.4 The Predictive Consistency Performance with Cohen's Kappa score

The result with %58 percent consistency shows both models nearly substantially fits with the labels of each other. This means they are both consistent enough to be used in analyses.

5.2.3 Sentiment Distribution Between Turkish and English

The results show that Turkish commentators prefer emotional expressions (anger %28, happiness %28) more than non emotional statements in their reviews. And English commentators prefer making neutral reviews more the other type of reviews. This shows Turkish people tends to show their feelings more than the English speaking people. Overall sentiment distribution is shown separately for both models in Figure 5.5.

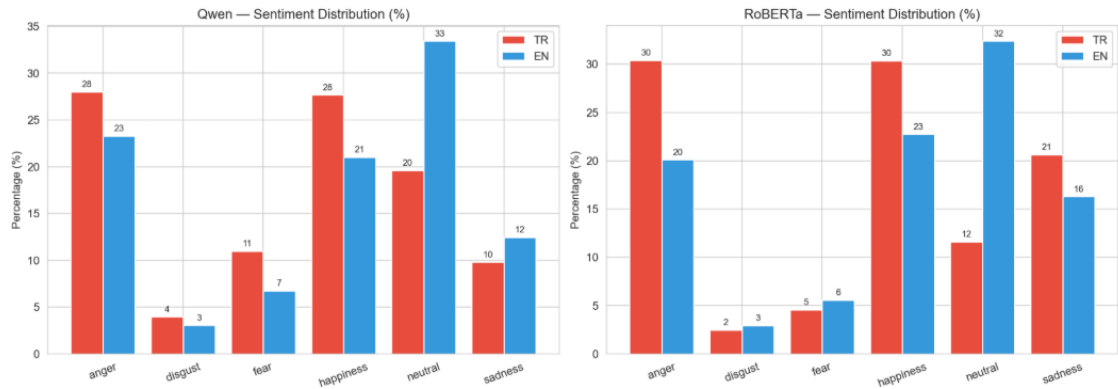


Figure 5.5 Sentiment Distribution Between Turkish and English Reviews

5.2.4 Category and Sentiment Analysis

For summarizing the observations at the subject level, the Turkish and English sentiment profiles are shown in Figure 5.8.

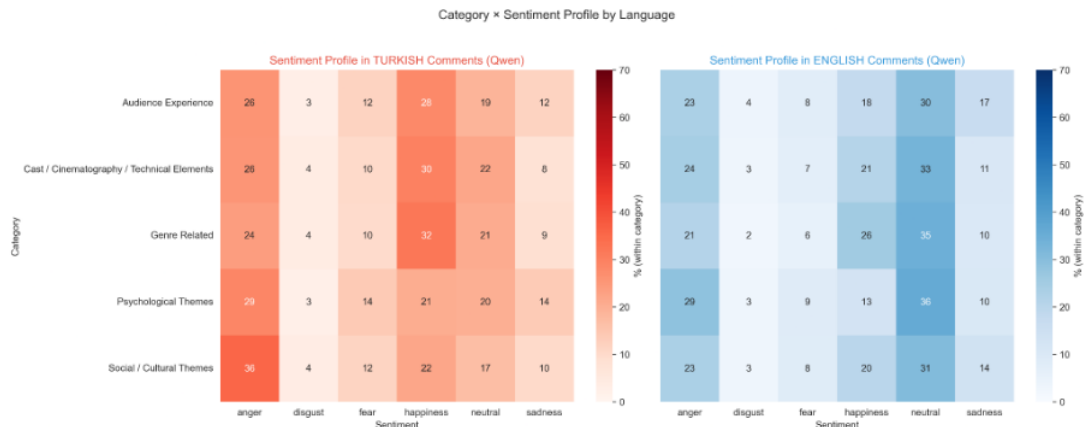


Figure 5.6 Category and Sentiment Profiles for Both Languages

The difference between Turkish and English reviews in the same category is shown in Figure 5.9.

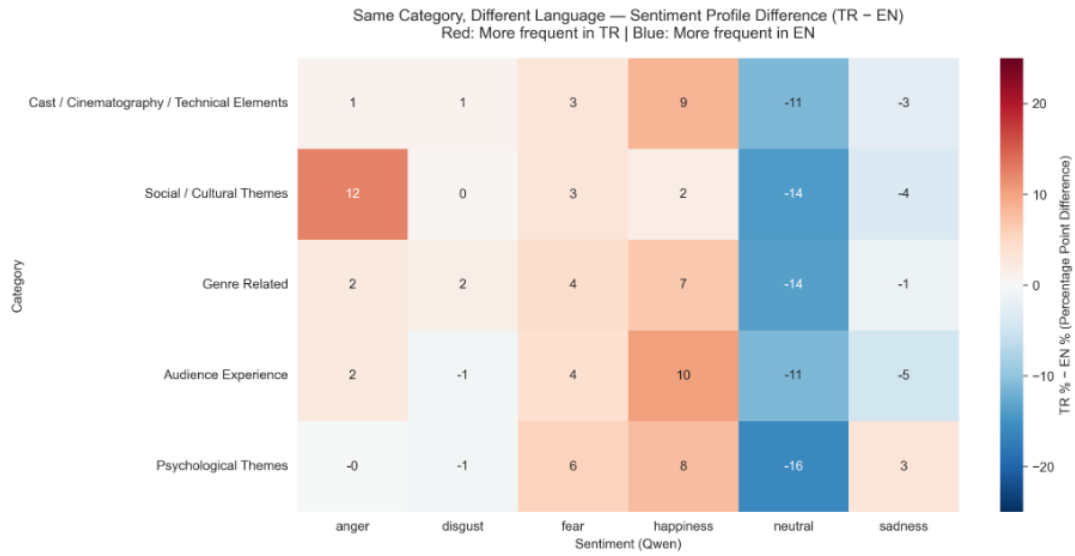


Figure 5.7 Sentiment Profile Difference Between Both Languages

The same asymmetry can also be seen in all 5 categories. In Turkish reviews there are higher rates of anger, fear, happiness in all 5 categories, while in English reviews there are higher rates of neutral and sadness comments. This results show that Turkish commentators are reflecting their happiness, grace, excitement, fear, anger, dissatisfactions, disappointments and their disagreements easier than English speaking commentators. And the graphs also show that English speaking commentators don't prefer this as much as Turkish speakers. English speakers are more likely to make reviews expressing their sadness and grief.

5.2.5 Topic and Sentiment Analysis

The Turkish and English sentiment profiles of each of the selected 23 topics are shown as a heat map in Figure 5.6. Each row represents one of the topics and each column represents one of the sentiment classes. The intensity of colours show how high is the percentage.

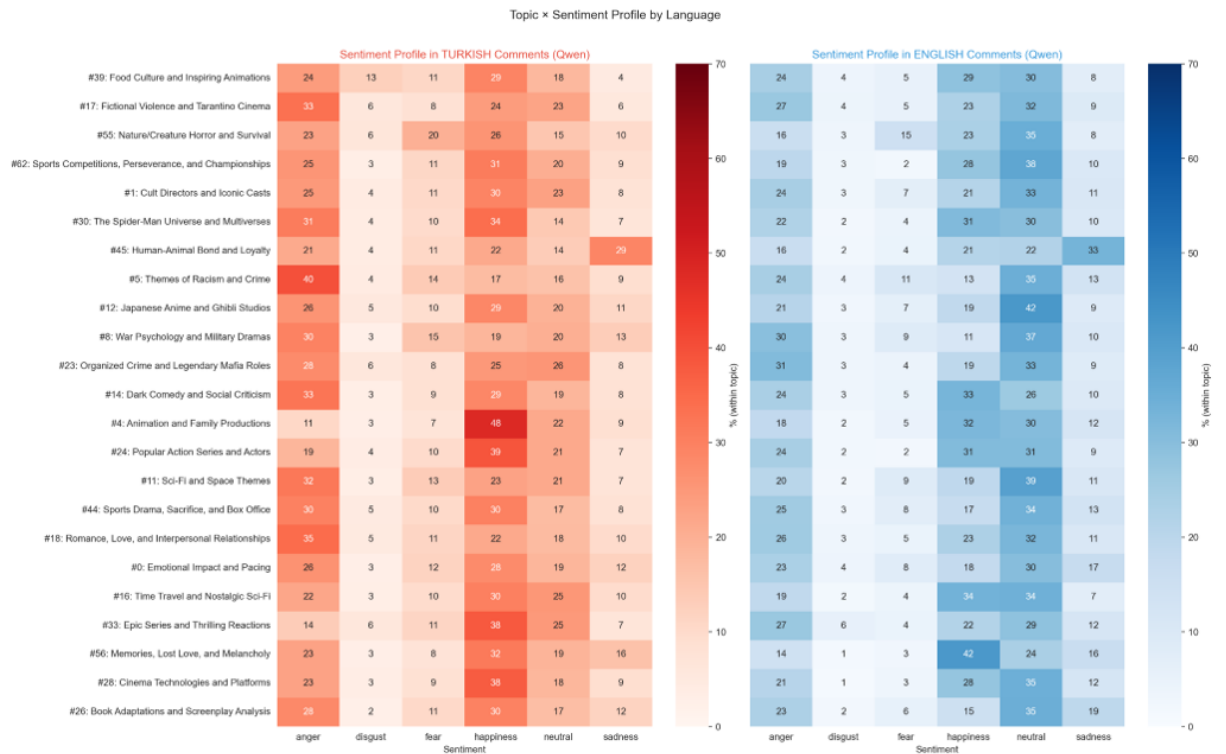


Figure 5.8 Sentiment profiles For Each Language

The differences in sentiment profiles between two languages on same topics are shown in Figure 5.7. Red cells indicate that those topics are more frequent in Turkish and blue cells indicates which topics are more frequent in English. Anger column being red and blue column being blue indicates that the cultural asymmetry can be seen consistently between the topics.

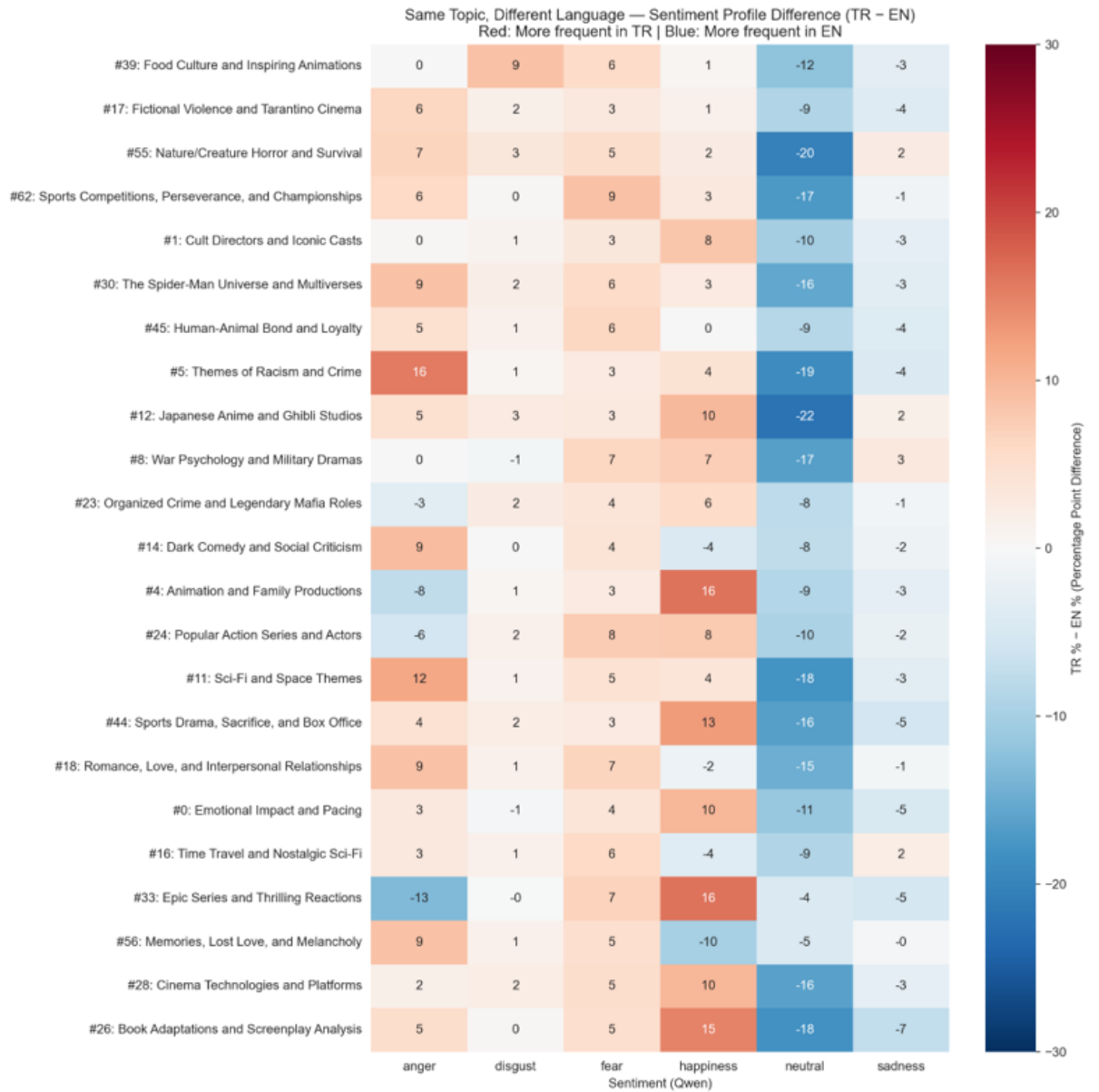


Figure 5.9 Sentiment Profiles on the Same Topics for Both Languages

5.2.6 Chi-Square Statistical Validation

The relationship between subjects and emotion classification was tested in this section as it was mentioned in 4.2.3. The test was made with 23 selected subjects and the emotion classification output of Qwen model. H_0 means subject assignments and emotion classification are independent. And H_1 means certain subjects are related with certain emotions. The test results are shown in the Table 5.4.

Table 5.4 Chi-Square Test Results (Topics and Sentiments)

Metric	Value
Contingency Table Size	23 x 6
Total observation	211,223
χ^2	7,648.39
dof	110
p-value	< 0,001
Cramer's V	< 0.085
Decision	< H_0 rejected

Since the p-value is quite below from 0.05, H_0 is rejected. This means emotion the distribution isn't random. Certain subjects are systematically related with specific emotions. This result confirms the project's main hypothesis. The semantic clusters created with BERTopic closely match with the emotional results created by Qwen.

The observed frequencies in the Contingency table show more significant deviations than expected results. For example, in Topic 5 (Crime/Detective), the anger category was nearly %24 higher than the expected (3,737 observed versus 3,016 expected). And in Topic 4 (Animation/Pixar) the happiness category was nearly %60 higher than the expected (5,276 observed versus 3,304 expected). This pattern is spread to all of the table and this means, the emotional tone of the topic theme is structurally determined rather than being randomly.

A Cramer's $V = 0,085$ value falls into the small effect class on the classical scale. The reason for this is the χ^2 statistic is directly sensitive to sample size. In 211,223 observations, even the small percentage of deviations can produce large χ^2 values. And this means, the p-value result proves that the relationship is statistically true. A small Cramer's V shows that not all reviews of a topic were put into the same emotion class. However, the emotion distribution profile varies structurally in different topics. This test confirms that both the BERTopic algorithm creates meaningful semantic clusters and the Qwen model makes consistent emotion predictions with these topics.

In this final section; the successful aspects, limitations are discussed first, and then suggestions for future studies are discussed later.

6.1 Successful Aspects

This study has achieved tangible and measurable gains in two areas.

Firstly, the $p < 0.001$ result in chi-square test numerically proves that the topics created with BERTopic and the labels generated by the sentiment classifier models are not coincidental. This is a direct statistical answer for the main question of this project. The topics in movie reviews are related with the sentiments of the reviews. This relationship is embedded to the data because of the nature of human behaviour.

Secondly, both of the independent models (RoBERTa and Qwen 2.5 7B) are showed that they are fit with the $\kappa = 0.462$ result. This shows that the findings of this project aren't caused because of the bias of a single model. The classifiers pointed to the same results, even though they were trained on different architectures, with different numbers of parameters and with different data approaches. This situation strengthens the validity of this project's results. And Qwen model achieved similar results with RoBERTa in both Turkish and English language. This shows that the large-parameter generative models can be successfully used in classification tasks with fine-tuning.

6.2 Limitations

The most significant limitation of the study is there is no manually labeled datasets about movie reviews for fine-tuning and testing the models. Even if the test set performance of the models are high in the used datasets, they can't be tested with the very large amount of labeled movie reviews. This limitation can be seen as the models are more likely to choose happiness and anger classes in the Turkish reviews.

And neutral class for the English reviews. So, it isn't possible to understand which model is more successful for this task without labeling really high amount of movie reviews by hand.

The second limitation is caused because of the used datasets aren't fitting with movie reviews. Since both of the TREMO and GoEmotion datasets aren't created with movie reviews, the maximum potential of the models are always limited.

Thirdly, Cramer's $V = 0.085$ falls into the small effect size category on the classical scale. The χ^2 statistic is directly sensitive to sample size. In 211,223 observations, even a deviation with small percentage can make a really high χ^2 value. This means, the subject-emotion relationship is statistically precise ($p < 0.001$) but it tends to follow the trends. All the reviews from a topic weren't put in a same emotion class, but the emotion distribution profile is differentiated with different topics.

6.3 Suggestions for Future Studies

Firstly, fine-tuning the models with datasets which contain only labeled movie reviews. When manually labeled sentences are directly given into the models, the rate of anger in Turkish comments and the rate of neutral comments in English may decrease. And extending this approach with self training may reduce the time and cost of labeling sentences. Even it can create an iterative adaptation for movie reviews.

Secondly, with assigning an LLM based model to representation_model parameter of BERTopic, titles and categories can be automatically generated for each topic. This will eliminate a manual part in this study and make the system more automated.

When these steps are applied together, the structural validity of current findings will be preserved and tendencies to specific classes may change for greater success.

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Project System Informations

System and Software: Windows İşletim Sistemi, Python

Required RAM: 16GB

Required Disk: 512GB